What would someone like me do with a tiny modular synth?

2nd Edition

Jess Borders playing a Workshop System at the Dyski Sound Maps residency in Cornwall in April 2024. Find how to join yourself at dyski.co

The Music Thing Modular Workshop System is a complete modular synth. It is slightly smaller than a hardback book. It comes in a foam-lined hard case. It comes as a kit; **you have to solder it together yourself***. Once built, doesn't make any sound at all until you patch it together.
It goes deep.

*We can also do this for you

What would someone like me do with a tiny modular synth?

Most musicians, whether they're writing songs, composing for film, making beats or basslines or noise, chase two things.

We want to create something new. And we want to make something that feels like us, personal and original. But we often get stuck.

One reason is convention; the ways instruments tell us they should be played. Play a 909 and a 303, sound like Daft Punk. Play an MPC, try to make boom bap.

These conventions do work. They're why genres sell and why instruments feel intuitive. But they're also limits to our imagination.

Escaping those conventions is hard. It requires confidence and stamina to make things that most other people won't understand or like.

The other reason is habit; the things our hands do automatically. When I pick up a guitar, the same chord progressions always appear under my fingers. I've been playing them for years and I'm bored of them. I open Ableton, the drums always end up in the same places.

The Workshop System is designed to help you escape both conventions and habits.

It's not just a synthesiser in the traditional sense of making basslines and leads. It's more like a toolkit for sound-making, a collection of things that make sounds and change sounds.

There's no menu-diving and no screens. Just knobs, switches, patch cables. It's a bit like freestyle Lego: you can't really tell what's going to emerge from the bricks when you start snapping things together.

You can make conventional instruments; a thick-sounding monosynth, a gnarly distortion box, a stereo filter.

But often, weirdness emerges. Take a field recording, or the Backyard Rain program card. Instead of using it as a texture, what if it becomes the lead?

Run it through the two filters to pull out peaks. Break the loop into pulses, into rhythm.

This is modular thinking; a little step outside the world, away from screens and goals and trying to do the right thing.

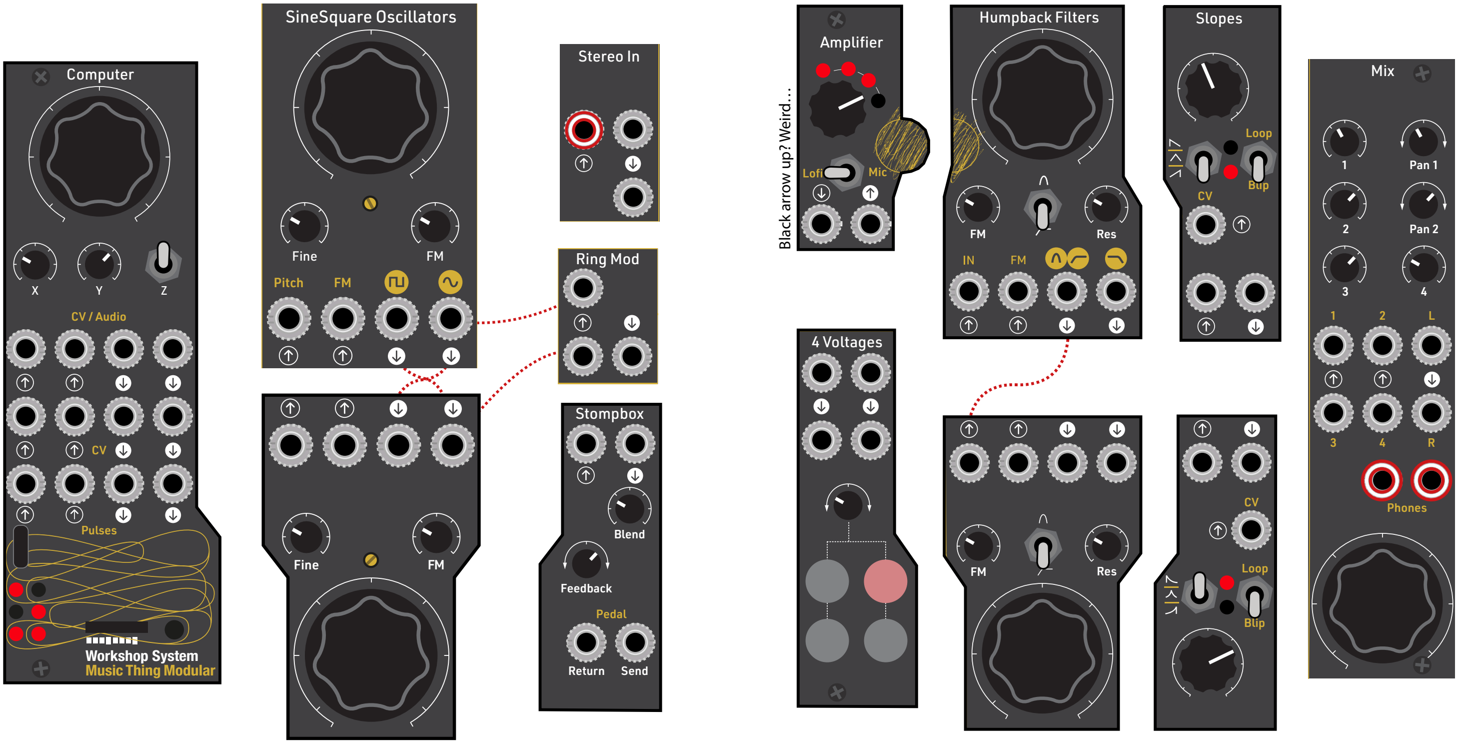
In modular there is no right, there is no wrong. There's no interesting and no boring. As John Cage said: "If something is boring after two minutes, try it for four".

Instead, there is just listening and responding, while the machine tries to make sense of you.

Tom Whitwell, London, 2026

What is it?

The Workshop System is 13 independent modules.
To make any sound, you need to patch them up.



Inputs
On the left
Black circle
White arrow up

Outputs
On the right
White circle
Black arrow down

Power The USB-C 15V input behind Stereo In

Switch Power on/off is behind Amplifier

Pedal Power 9v power out is behind Stompbox

Normalised An internal patch cable connects these sockets

Stereo These sockets are stereo, all others are mono

SineSquare Oscillators
Two simple analogue circuits, just a handful of transistors and chips, evolved from 70s designs by Bernie Hutchins. Optimised for FM, they just sound nice, and go from melodic to CCK-KNNEERRK very quickly.

Stereo In
The Workshop System is made to connect to stuff you already have. Plug in your laptop, phone, drum machine, Volca, this boosts the signal to modular level.

Amplifier
Behind the panel is a contact mic to pick up physical sounds: switches, touches, thumps. Also a nice transistor distortion based on the Mini Moog.

Humpback Filters
A pair of old-tech filters, designed by God's Box. Low-pass and high/bandpass outputs, clean when you want them clean, gnarly when you want them gnarly.

Ring Mod
is a great-sounding ring modulator, also works as a simple VCA. Turns sine waves into bell sounds, drum loops into $%@!.

Computer
The heart of the Workshop System is a sound computer that runs on tiny program cards, each one a distinct firmware in physical form. For example, one card is a Turing Machine random looping sequencer, great for unpredictable melodies. Another is a great-sounding 90s-style Reverb. Another, a USB MIDI interface. Just swap cards, hit reset, and the module completely changes its behavior.

Mix
Is a simple output mixer, with two stereo channels, two mono channels and a powerful headphone amp with two outputs.

Power Supply
The System runs on USB-C PD or 15-22v barrel connectors. Use battery powerbanks and laptop adaptors, most likely something you've already got at home.

Making a sound
To make any sound at all, you need patch cables. Patch a Sine wave to the mix input. Wiggle the top knob on the oscillator and the sound should wiggle.
This simple patch is a complete instrument: one pure sine with a big knob to control pitch, and another to control volume.
A theremin, with the interface of an Etch-a-Sketch.
So stop here, master this instrument and use it to make a lot of music that you enjoy.

Stompbox
Connects guitar pedals, with a feedback control to make boring pedals more fun. Plus a 9v power out for pedals behind this module.

Voltages
A *minimum viable keyboard*: four voltage outputs, one knob, four buttons. Patch, push, change. More fun to play with than to explain.

Slopes
Use this to create versatile LFOs, envelopes, clocks and pulses. It's very loosely based on the Serge Voltage Controlled Slope circuit.

Patch 1: Keep it simple

CABLES: 1
CONTROLS: 2
CHAOS POTENTIAL: LOW

This is the simplest possible patch on the Workshop System. I use it to reset the system and my ears each time I start.

Set up the top SineSquare Osc: put the main knob at about 12 o'clock and the FM knob down.

Set up Mix set the top two little knobs (Channel 1 and Channel 1 Pan) to about 12 o'clock.

Connect the headphone outputs to something: Headphones, a speaker, a mixer.

Connect one patch cable: from the top Sine ∿ output to the Mix 1 input.

Gradually turn up the volume: you should hear a sound. Move the frequency knob if you can't hear anything yet.

This patch is a perfect minimal electronic instrument: one sine wave with one big knob to control its pitch, and a second big knob to control volume. It's a theremin, with the interface of an Etch-a-Sketch.

In a perfect world, you'd stop here, master this instrument and use it to make a lot of music that you enjoy. Only then, when you've exhausted this instrument completely, should you move on and try a third control or even a second patch cable.

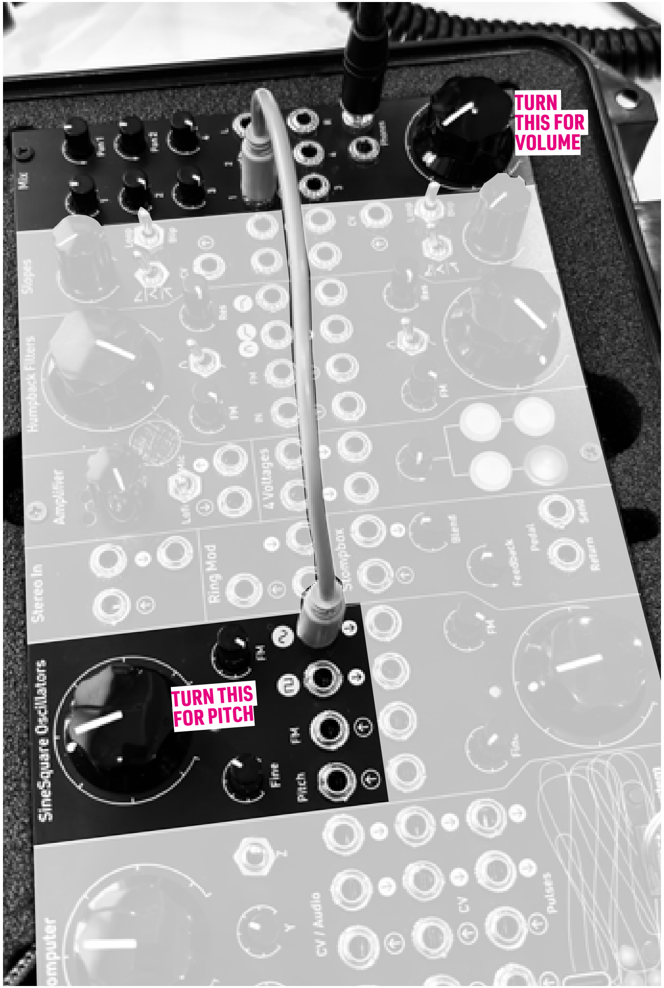
That's what we did - for a couple of hours at least - at the first Dyski retreat in Cornwall.

Everyone recorded one minute sketches using this patch, using single-line pencil drawings as the score. After an hour, we listened back to people's recordings, which were… wobbly but interesting and varied.

Then we spent another hour doing multitrack versions of the same thing. These results were incredible: rhythmic pulses, curtains of sound, drama, tension, surprise.

You can hear those recordings on the Dyski album "Port Navas Sessions: Sine Wave Portraits.'

Spend some time with this patch before moving on.



Patch 2: Make it weird

CABLES: 1
CONTROLS: 5
CHAOS POTENTIAL: HIGH

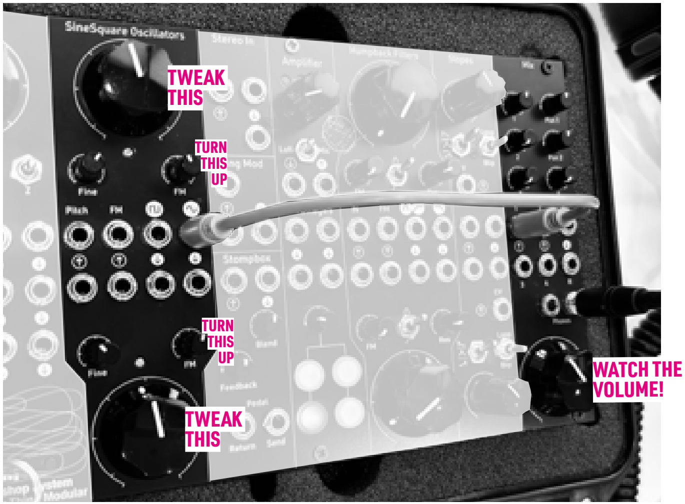
Turn up the FM: Without changing any cables, gently turn up the little top FM knob.

FM stands for Frequency Modulation. It means that the bottom oscillator changes the frequency of the top oscillator.

The signal is normalised; it goes through one of those hidden patch cables.

Now you have five knobs to play with. Frequency and volume like before, but now FM and the bottom oscillator's big frequency knob change the sound.

Now turn up the bottom FM control. Bottom influences top, top influences bottom. It's chaotic. Small changes have big effects. Listen carefully, move slowly. Can you control what's happening?



Patch 3: Filters!

CABLES: 2
CONTROLS: 4
CHAOS POTENTIAL: MEDIUM

Send a square wave to the filter: one cable from to IN on the top filter, and turn off FM.
Choose a Low Pass: Patch from low pass on the top filter to Input 1 on Mix. Turn Res down.
Tweak the big knobs on the oscillator and the filter. What do you hear? Try to find the edges. Turn the oscillator right down, so it's just clicks. What effect does the filter have on the sound of those clicks?

Gently turn up the Res (Resonance). This makes the filter effect more extreme.

There's lots ot experiment with here. Move from low pass to band pass / high pass and flip the switch. What do you hear?

Patch a sine from an oscillator to FM on the top filter and gradually turn up the FM control to see what that does.

Move the output cable to the bottom socket to find out what the hidden patch cable between the filters does.

What can I do with the Computer?

The Computer module was inspired by the early music studios like EMS in London and Bell Labs where computers were used to control analogue oscillators and filters, compose algorithmic music and generate waveforms.

Computer runs on little program cards. You get four with the kit: a Reverb, a MIDI interface, a Turing Machine sequencer and a blank.

What can you put on the blank? By far the most important part of Computer is the developer community around it, who create WILD program cards; deep programming environments, MIDI polysynths, a keyboard-controlled speech synth, a laser interface, a feedback system inspired by a Dutch composer from the 60s, noise generators,a Benjolin, drones and signal processors and drum machines and something new and bonkers every week.

The cards on the right are just a sample of what is out there. Some are for sale, like RYK's suite of incredible cards, and many are available free from the Computer github.

Find out more and join the community:
musicthing.co.uk/discord
musicthing.co.uk/Computer_Program_Cards

20 Reverb+
Versatile 90s-style plate reverb that comes with a MIDI interface and sequencer.

21 Resonator
Lush melodic tones, good for drone, chords as well as plucking sounds.

03 Turing Machine
4 x random sequencers, with note, pulse and modulation outputs.

41 Blackbird
Super deep scripting and live coding environment, based on monome crow.

42 Backyard Rain
Crossfaded field recordings from a Seattle backyard. Why not?

25 Utility Pair
24 incredibly useful small utilities, from delays to wavefolders. Use 2 at once.

22 Sheep
A granular processor and digital degradation playground with quirky controls

910 Pitched Delay
From RYK, an awesome 70s delay for Bowie & Laurie Anderson vibes

B77 Tape
From RYK an intuitive varispeed tape style 90 second two track audio looper.

Where can I learn more?

There's a deep community around the Workshop System and Tom's monthly newsletter is the best way to keep up:
musicthing.co.uk/newsletter
You can also jump right in and join the Discord
musicthing.co.uk/discord

Can I try a Workshop System?

You can play a system at SchneidersKeller on Denmark Street in London, or play one for freelin the browser:
musicthing.co.uk/simulator

How do events happen?

To attend an event, sign up to the newsletter above and look out for events near you. If you have an idea for an event or want to borrow some Workshop Systems, email Tom:
tom@musicthing.co.uk

Can I really build one myself?

Most people can build the Workshop System in an evening or two. It's not too hard, there is no SMD, but if you've never soldered before, practise on something else first.
Watch the real-time build video, the closest thing you can get to an online build workshop.
musicthing.co.uk/buildvideo

Can I buy one right now?

Complete DIY kits are available from Thonk, Exploding Shed and Perfect Circuit.
Fully assembled systems are available from Signal Sounds, SchneidersLaden, SchneidersKeller and Perfect Circuit.
To find your nearest dealer, contact **support@thonk.co.uk**

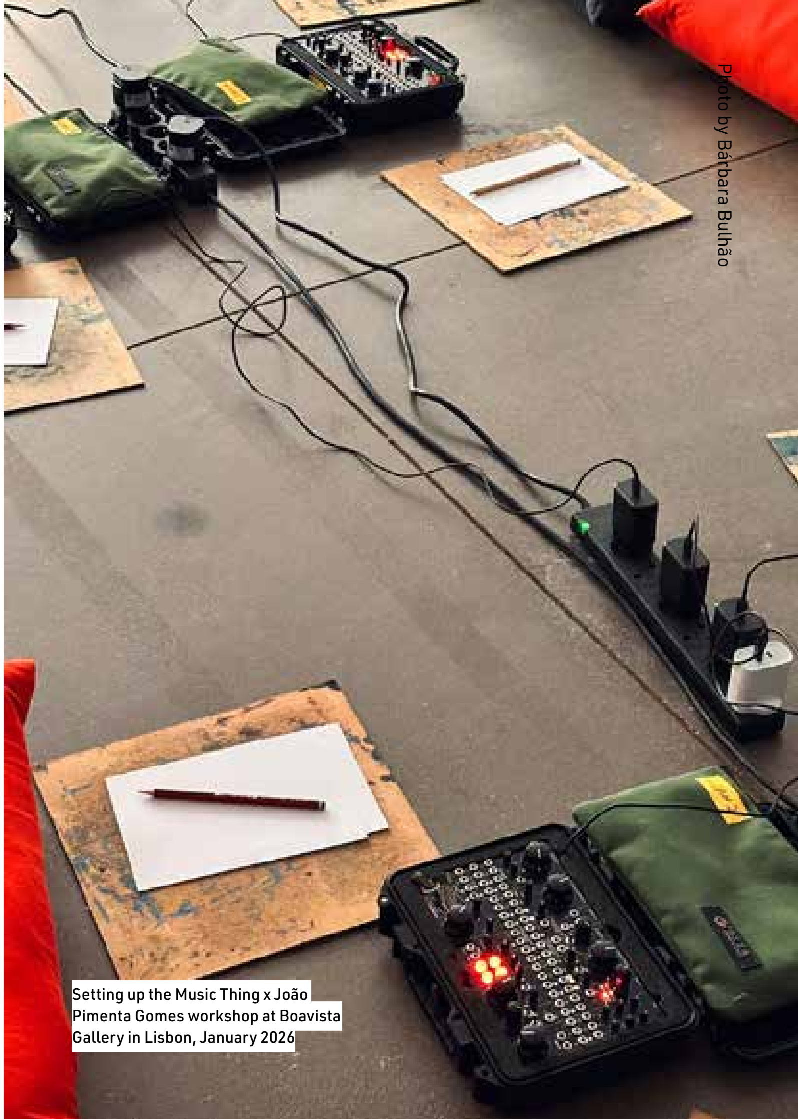


Photo by Bárbara Bulhão

Setting up the Music Thing x João Pimenta Gomes workshop at Boavista Gallery in Lisbon, January 2026

"Fabulous in almost every way. A triumph of a compact system, the Workshop System is powerful and highly flexible – and is excellent value for money. Sound on Sound Magazine

"This project is... wow." hawksquill

"What a beautiful little box." tekknovator

"Limitations breed creativity! Deep knowledge of a small set of tools is much more inspiring that all the gear and no idea!" Matt

"I have been wanting to get into synthesis for decades now, but this has pushed me to jump in, what an unbelievably great piece of kit" Cloudwerx

"After 6 months, it was the best purchase I've ever made. I love it and have found it in me to start performing again" Luke Bowers

"The Workshop System kit has meaningfully enriched my life" Brian

"It's a wonderful system and I am having a lot of fun with it. I am still amazed that my terrible soldering has worked!." Fanzini